

Gabriel Guimarães Carvalho

Research Scientist | Computer Vision | Machine Learning | Ph.D. Physics

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Summary

Hi, I'm Gabriel. I'm a Brazilian mathematical physicist who enjoys turning mathematics, scientific computing, computer vision, and AI into intuitive solutions for understanding visual data. I have international experience from completing my PhD in Italy and gaining work experience in Belgium, where I now reside. I'm also fascinated by the future of computing, especially quantum technologies and generative AI.

Education

Università di Roma "La Sapienza" <i>Ph.D. in Physics and Relativistic Astrophysics</i>	2014 – 2017 Rome, IT
Universidade Federal de Pernambuco (UFPE) <i>M.Sc. in Mathematics</i>	2011 – 2013 Recife, BR
Universidade Federal de Pernambuco (UFPE) <i>B.Sc. in Mathematics</i>	2008 – 2010 Recife, BR

Experience

Research Scientist <i>imec</i> Deep Learning - Computer Vision - Genetic Algorithms - R&D in several domains	Jun. 2024 – Present Antwerp, BE	Postdoctoral Researcher <i>UFRPE</i> Alternative gravitational theories and geometric methods in Physics	2017 – 2018 Recife, BR
Senior Research Scientist <i>IDLab - UAntwerpen - imec</i> Deep Learning - Computer Vision - R&D	Jan. 2024 – May 2024 Antwerp, BE	Lecturer <i>UFPE</i> Calculus I, II, III - Analytic Geometry - Linear Algebra - Calculus w/ One Complex Variable	Nov. 2017 – Jul. 2019 Recife, BR
Computer Vision Engineer <i>SpaceTime Labs & Moray</i> Computer Vision - Image Processing - Data Preprocessing/Analysis - Hyperspectral Imaging - Geographic Information System (GIS)	Jul. 2021 – Nov. 2023 São Paulo, BR (Remote)	Lecturer <i>UFPE</i> Calculus I, II - Analytic Geometry	Apr. 2012 – Jan. 2014 Recife, BR
Ph.D. Researcher <i>Projeto CIn - Motorola</i> Computational Photography - Deep Learning - Data Science/Analysis - Image Processing - Image QA	Sep. 2019 – Jun. 2021 Recife, BR		

Skills

Languages: Proficient: Python. Familiar: Go, Matlab, C++, Java	DevOps: Git, Docker, AWS. Familiar: MLFlow, DVC
ML/CV: PyTorch, PyTorch Lightning, OpenCV, Scikit-Learn, Scikit-Image, NumPy, SciPy, Pandas	OS: Linux, macOS, Windows
Geospatial: QGIS, GeoPandas, Rasterio	Spoken: Portuguese (native), English (C1/C2), Italian (B2/C1), Norwegian (A1/A2)

Certifications

IBM DevOps & Software Engineering · Deep Learning (DeepLearning.AI) · Math for ML (Imperial College) · ML (Stanford)
Advanced CV with TensorFlow · Quantum Information



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